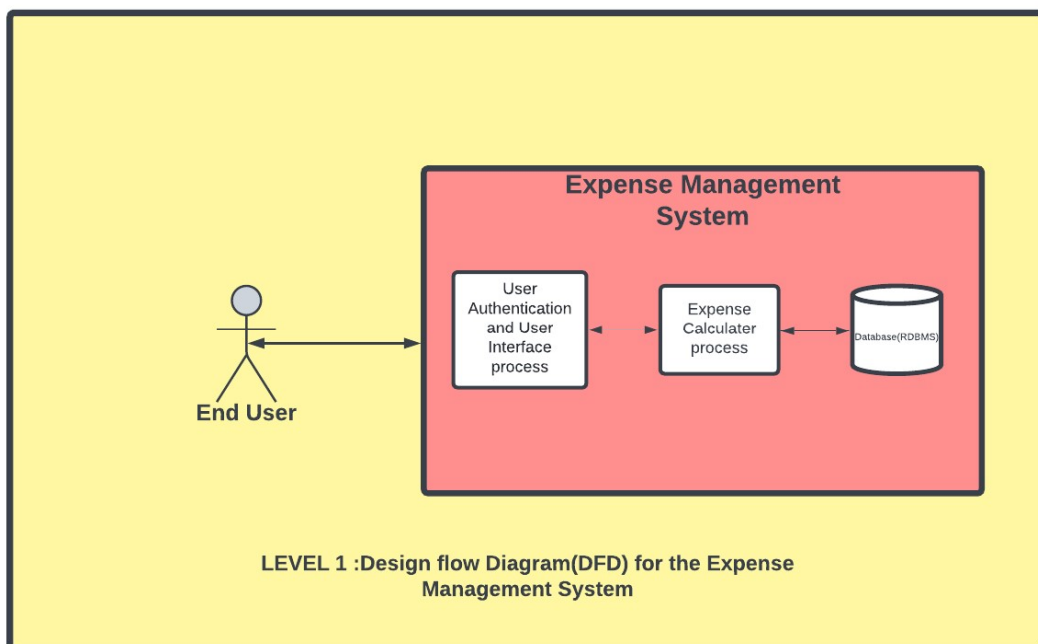
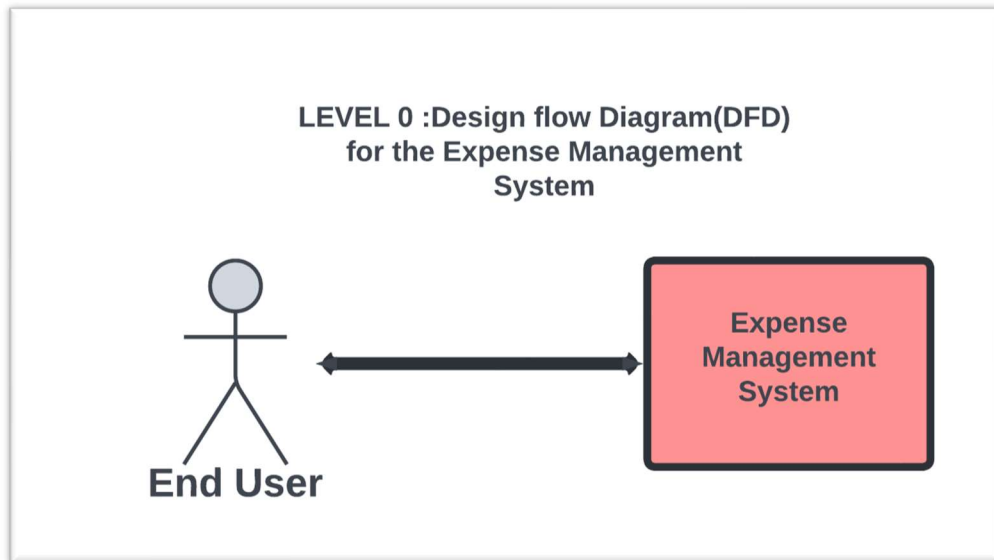


Section B: Design

High Level Data Flow Diagrams

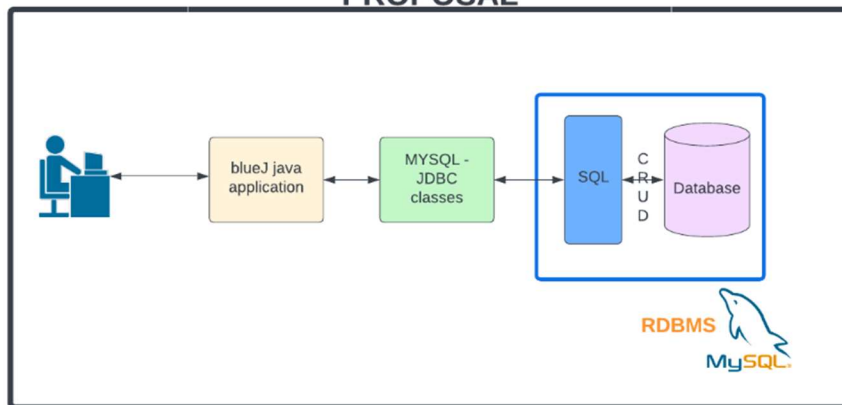


Expense File will be stored in a relational database as client needs multiple data table to be handled in future for comparison. Refer Appendix:

- DFDs after consultation with my client Mr Bhat (refer **Appendix A, B, C**)

Design Proposal:

EXPENSE MANAGEMENT SYSTEM [EMS] DESIGN PROPOSAL



SQL: Structured Query Language. SQL is a standard language for storing, manipulating and retrieving data in databases.

JDBC: Java Database Connectivity

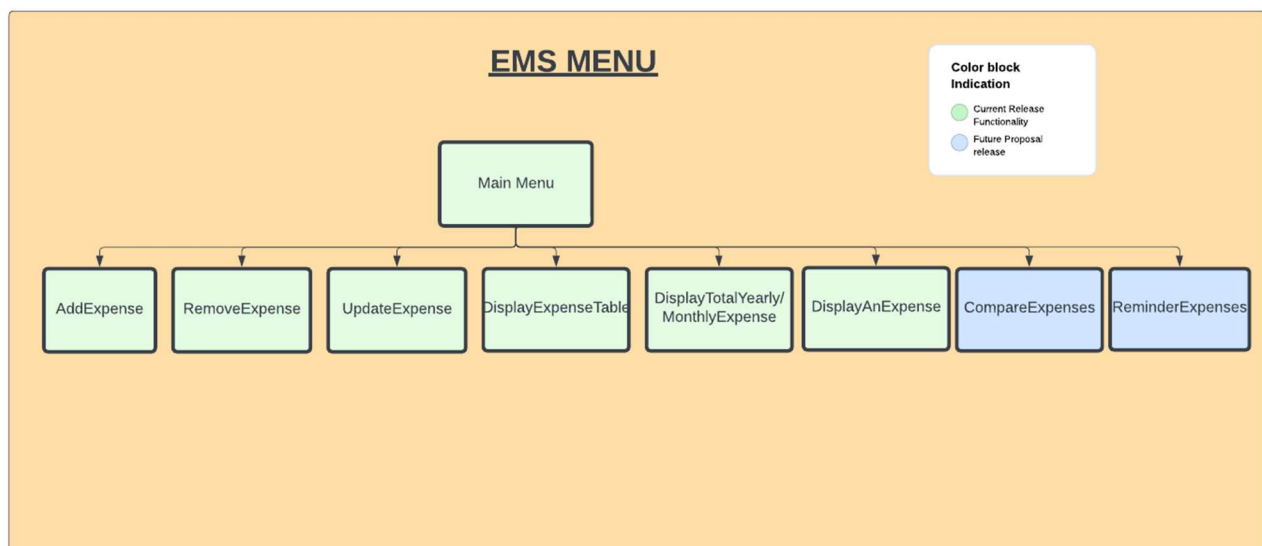
RDBMS: Relational DataBase Management Systems. It is basically a program that allows us to create, delete, and update a relational database

C : Create/Insert

R: Read /Select

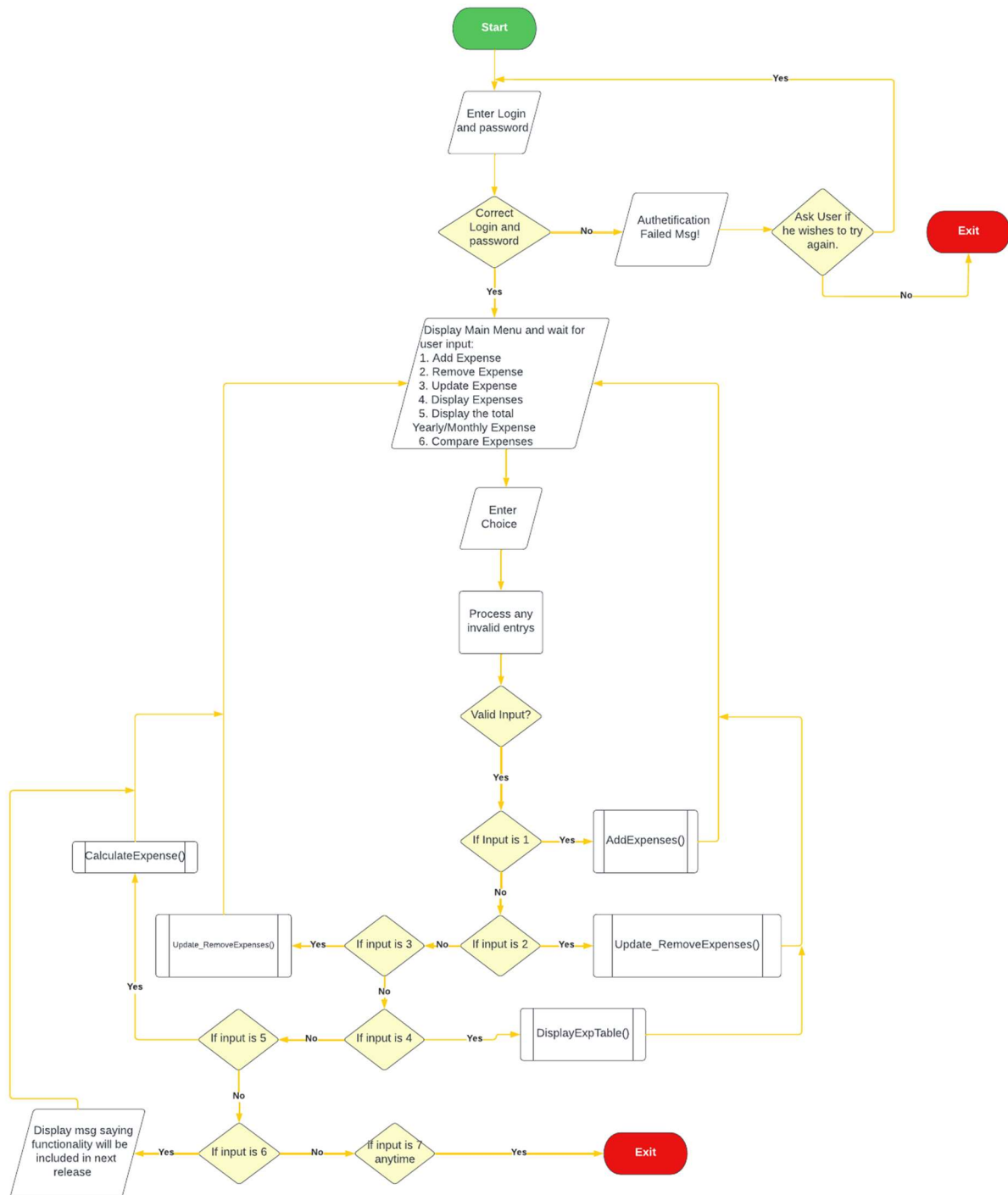
U: Update

D: Delete



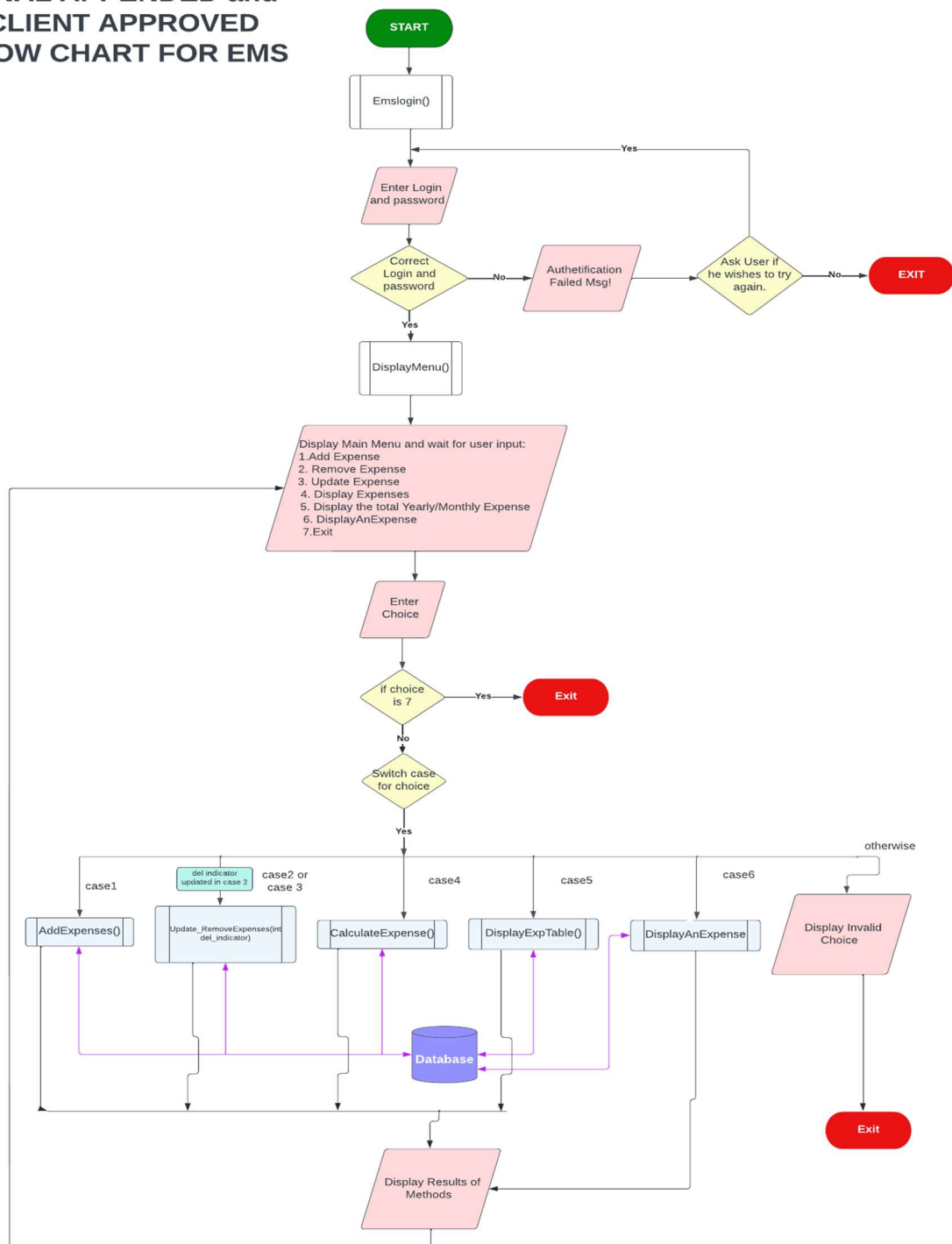
- DisplayAnExpense: added after client's request (refer **Appendix D ,E ,F**)
- CompareExpense , ReminderExpenses will be added in future release (refer **Appendix E,I**)

Flowcharts



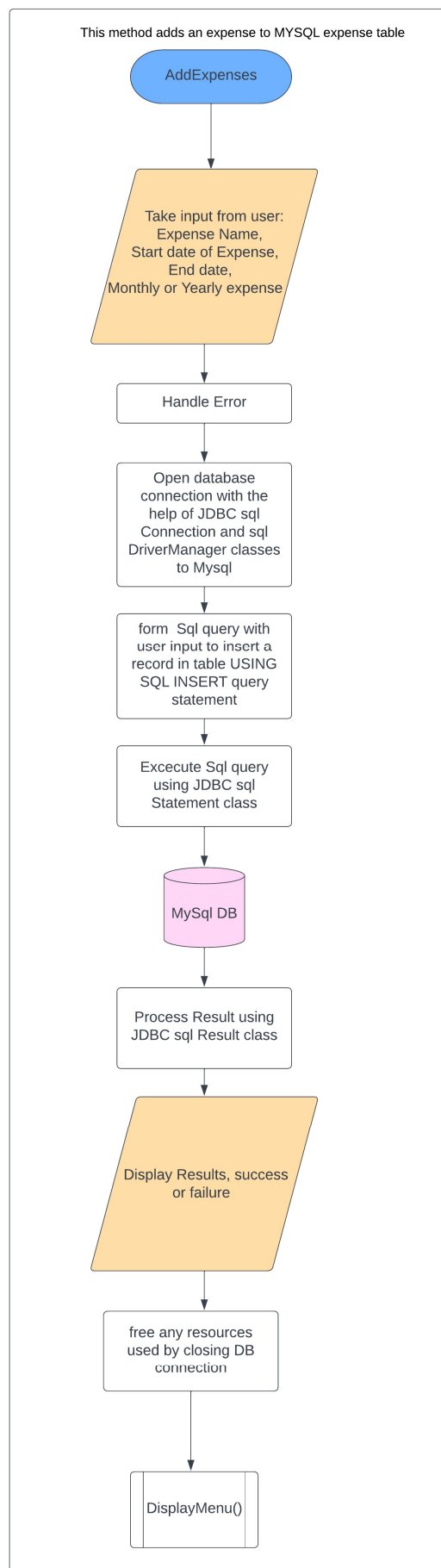
INITIAL FLOW CHART FOR EMS BEFORE CLIENT's REQUEST FOR CHANGE

FINAL APPENDED and CLIENT APPROVED FLOW CHART FOR EMS

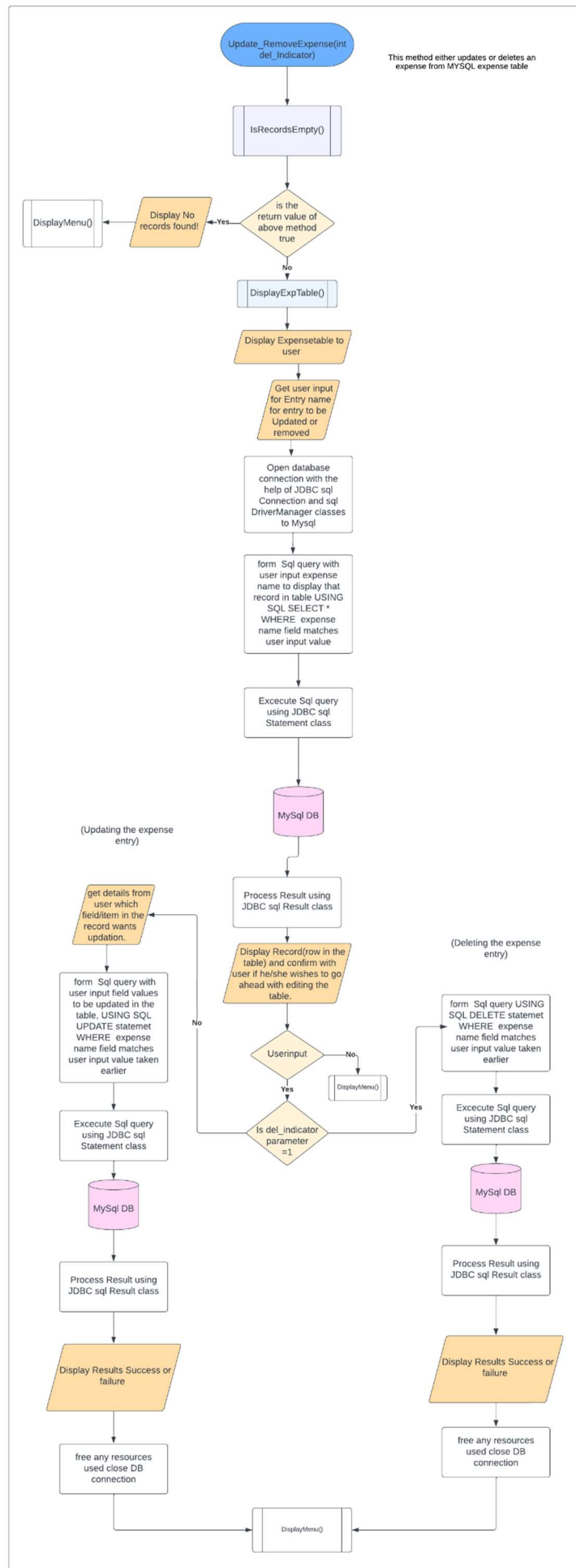


Flowcharts of some important methods:

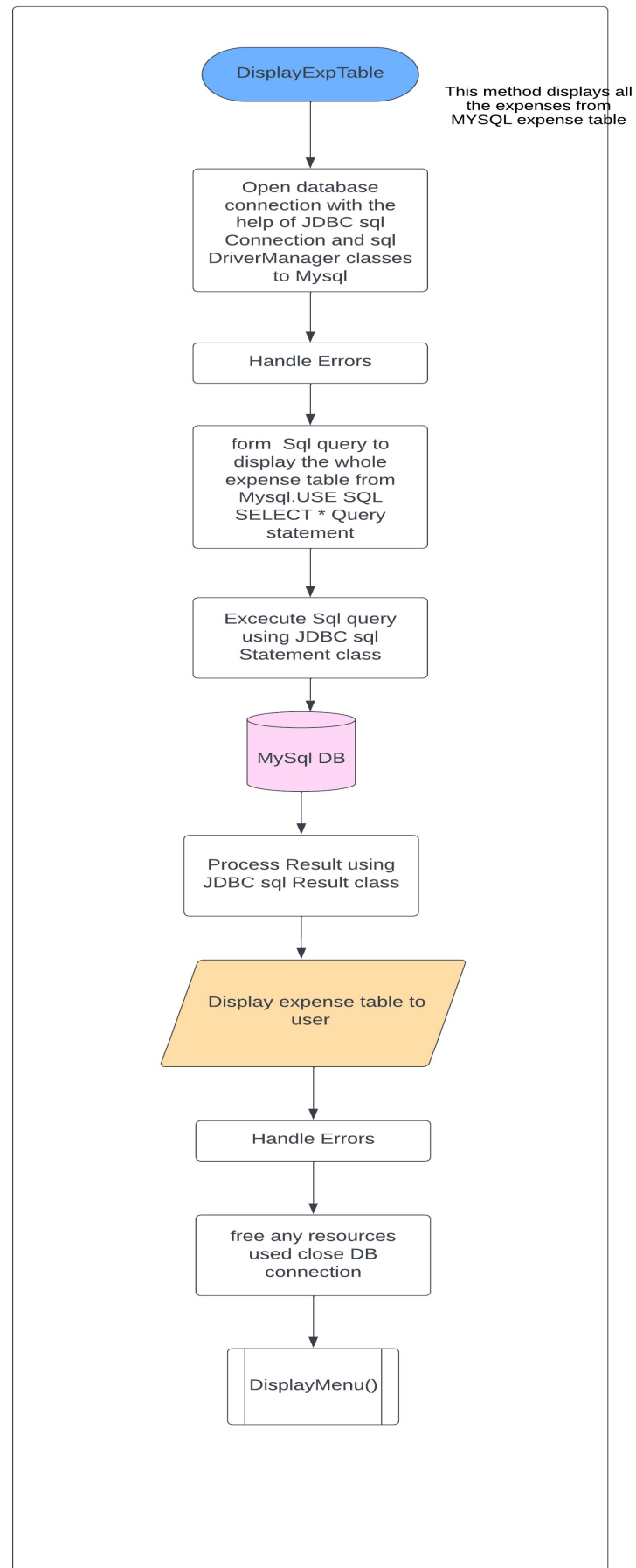
Flow Chart for Method :AddExpense



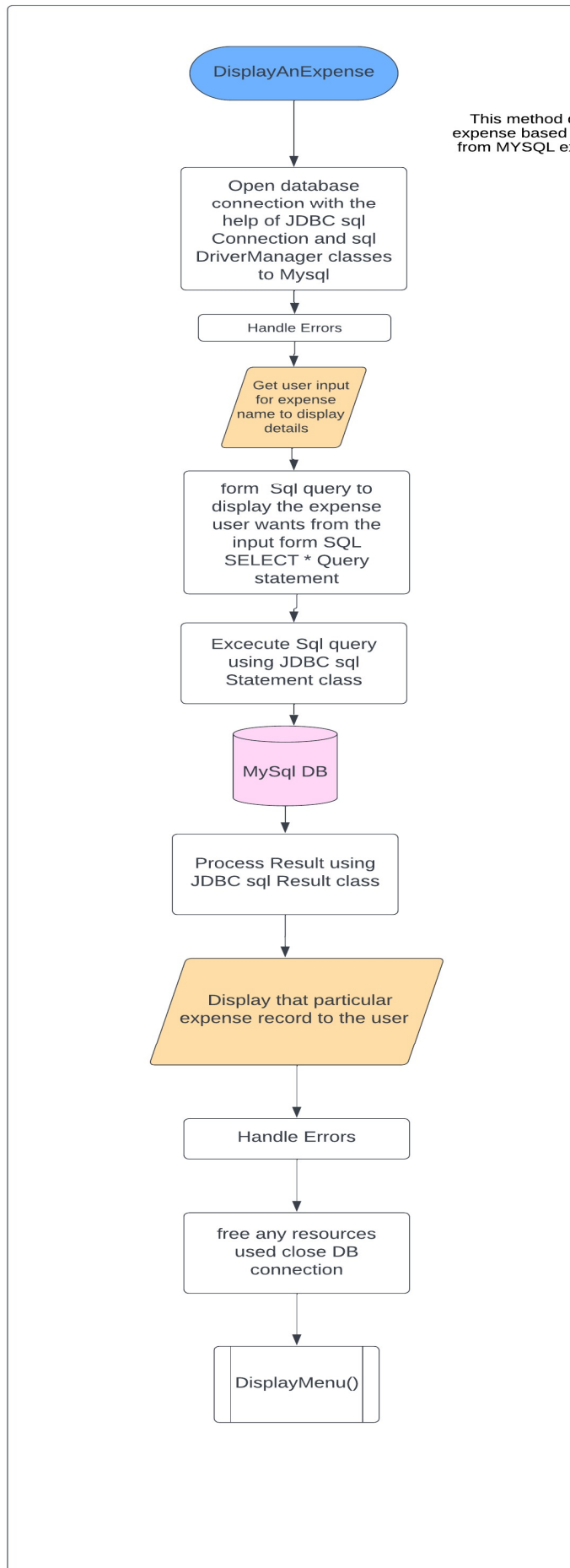
Flow Chart for Method Update_RemoveExpense



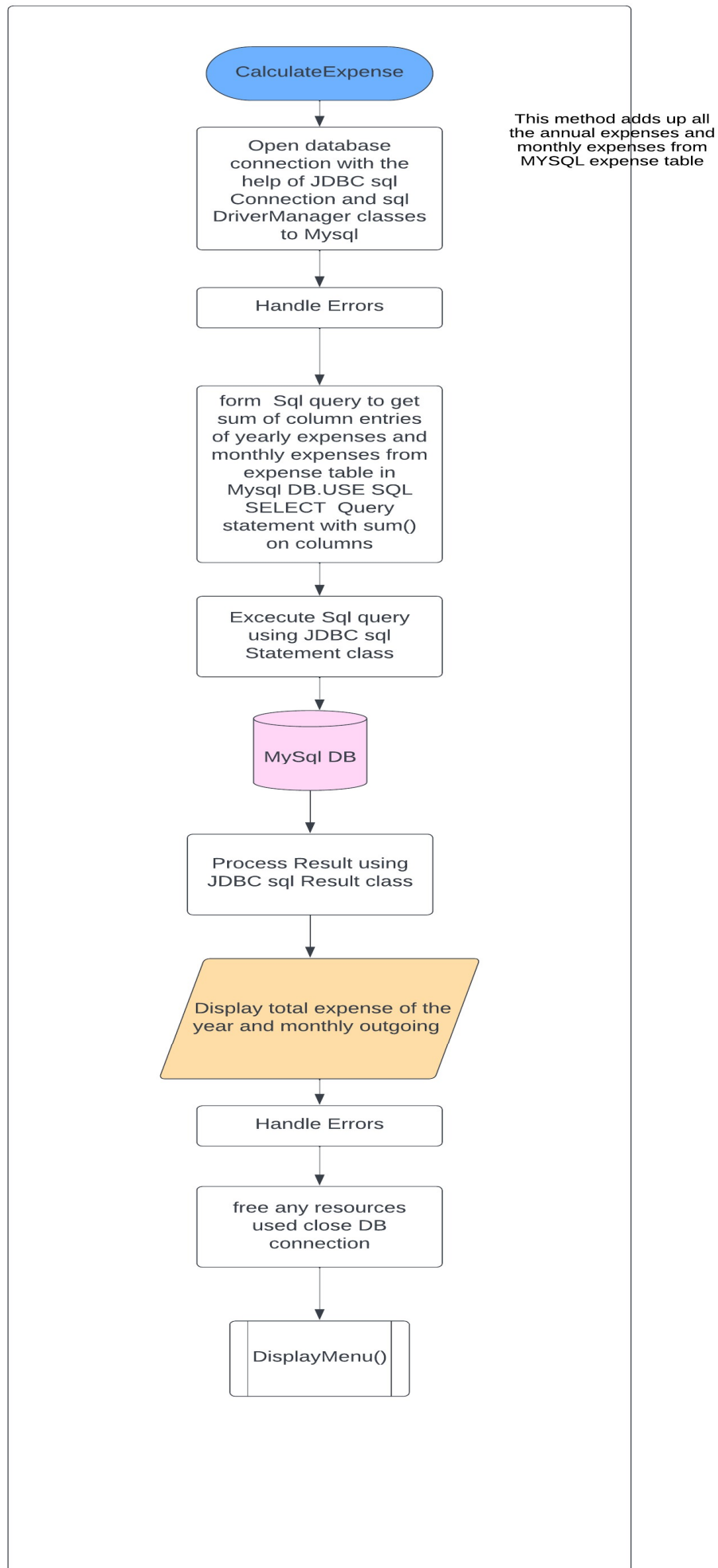
Flow Chart for Method :DisplayExpTable



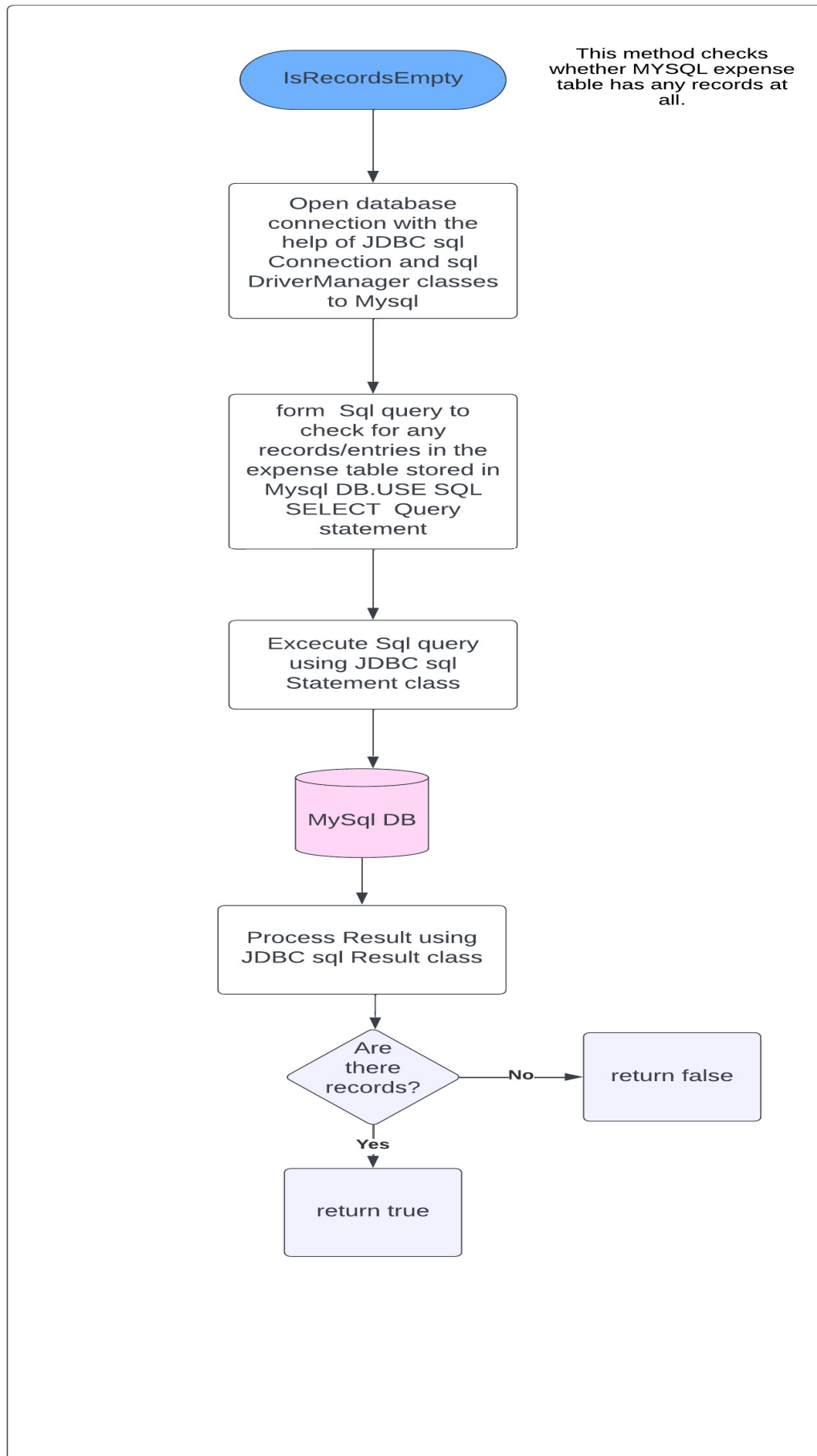
Flow Chart for Method :DisplayAnExpense



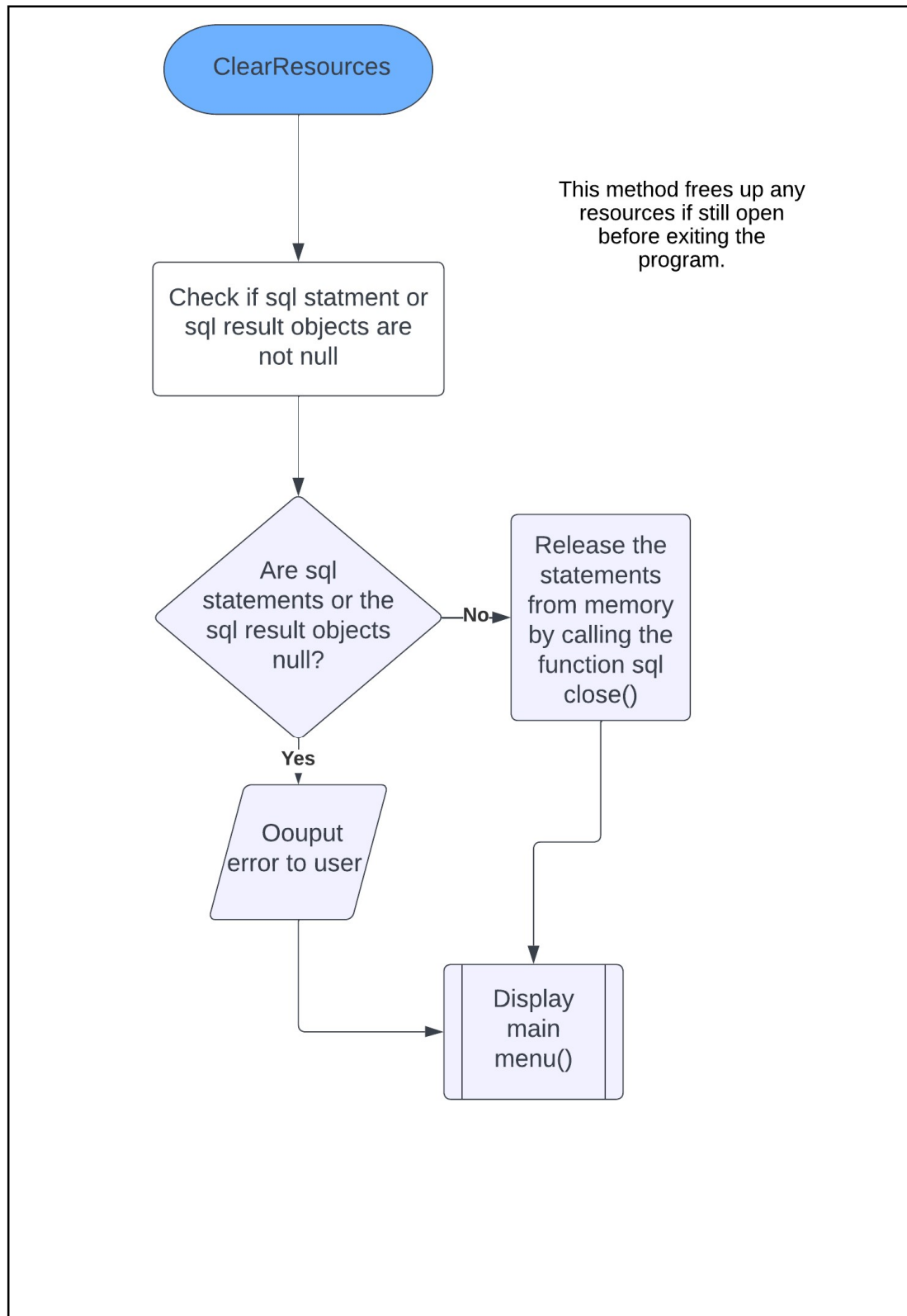
Flow Chart for Method :CalculateExpense



Flow Chart for Method :IsRecordsEmpty



Flow Chart for Method :ClearReasources



UML DIAGRAMS:

UML Class Diagram for EMS

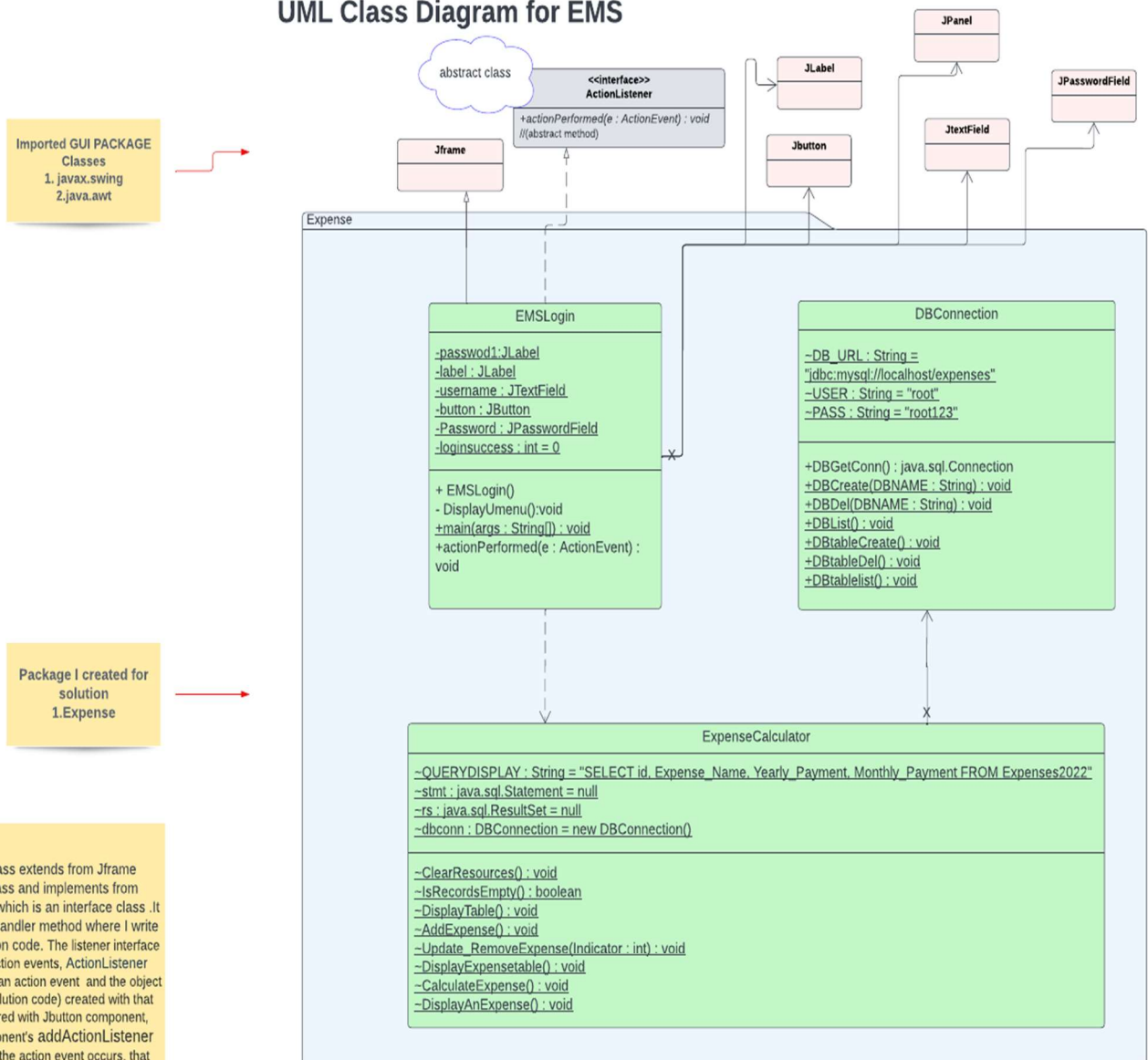
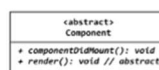
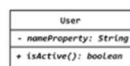


Fig 1



Abstract class

Same as the interface shape. Abstract methods marked as abstract with comments or "abstract methodName(): returnType".



Class

Properties or attributes sit at the top, methods or operations at the bottom + indicates public, - indicates private, and # indicates protected

UML HELP



Inheritance

B inherits from A. Creates an "is-a" relationship. A is a generalization.



Implementation/realization

B is a concrete implementation/realization of A.



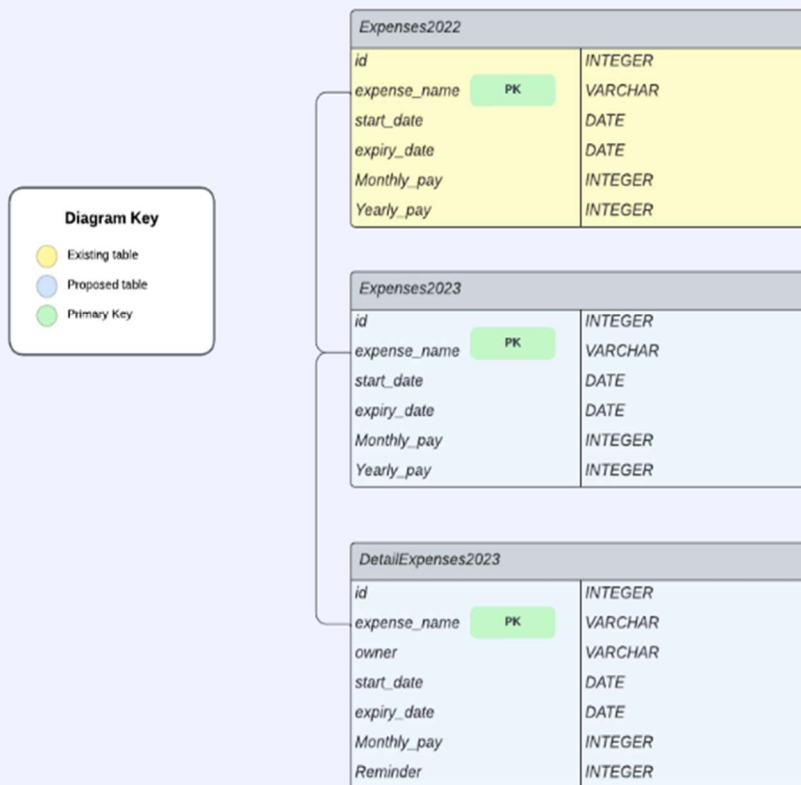
Dependency: A depends B, Changing B can foul up A. Often A uses some of B or A contains B objects, etc.



A4 is not navigable from B4 while B4 is navigable from A4.

Database Design:

MYSQL DATABASE: Expenses



- In future release the database will be handling multiple expense tables (in light blue) and will contain a *reminder* field which indicates if reminder for expiry should be set or not.
- We can specify primary key: helpful in queries handling multiple related data tables. This will be useful while *comparing* expenses from two different tables. (Refer Appendix I)

Graphical Visualization of Expense table Display for the user:

Expense Name	Start Date	Yearly Payment	Monthly Payment	Expiry Date
Broadband	2022-06-01	1200.0	100.0	2030-03-04
Mortgage	2022-01-01	180.0	15.0	2023-04-04
Council Tax	2022-01-01	2500.0	208.3	2023-01-01
Gas and electricity	2022-01-01	2400	200	2023-09-12

Main Menu / User Interface

Initial UI design:

```
-----  
---          MAIN MENU          -----  
-----  
  
Press 1 to Add Expense           -  
Press 2 to Remove Expense        -  
Press 3 to Update Expense        -  
Press 4 to Display Expense Table -  
Press 5 to Display Yearly/Monthly Expense -  
Press 6 to Compare Expense       -  
Press 7 to Quit
```

1

Please enter the Name of your Expense
Disney+ TV Subscription

Please enter the length of your expense in months:
12

Will you be paying this expense monthly or yearly? (enter m/y)
y

How much will you be paying yearly?
75.00

Please enter the start date for the expense contract:
31.07.22

The end date for this expense contract is: 31.07.23

Is this correct? (please confirm y/n)
y

The total amount of money to be paid for this contract: 75.00

Here are the expense details:
Expense Name: Disney+ TV Subscription
Start Date: 31.07.22
End Date: 31.07.23
Expense Duration: 12 months
Expense Cost: 75.00 per year
Total Amount To be Paid: 75.00

2

Which expense would you like to remove?

Disney+ TV Subscription

The expense has been removed.

[display expense list]

3

Please enter the Expense Name you would like to update:

Disney+ TV Subsc

Invalid Expense Name.

Would you like to re-enter your Expense Name?

yes

Disney+ TV Subscription

Which feature would you like to update in your Expense List:

1. Expense Name
2. Expense Duration
3. Expense Cost
4. Start Date
5. End Date

2

[Update Expense Duration]

[Display new Expense List]

4

Which Expense Name would you like to be displayed?

Disney+ TV Subscription

Here are the expense details:

Expense Name: Disney+ TV Subscription

Start Date: 31.07.22

End Date: 31.07.23

Expense Duration: 12 months

Expense Cost: 75.00 per year

Total Amount To be Paid: 75.00

5

[Display the Total Expense Yearly]

6

Which two Expenses would you like to compare?

Disney+ TV Subscription

Broadband

Expense Name: Disney+ TV Subscription

Start Date: 31.07.22

End Date: 31.07.23

Expense Duration: 12 months

Expense Cost: 75.00 per year

Total Amount To be Paid: 75.00

Expense Name: Broadband

Start Date: 11.10.21

End Date: 11.08.23

Expense Duration: 22months

Expense Cost: 30.00 per month

Total Amount To be Paid: 660.00

Updated Menu UI design after Client Request (Reference Appendix E,H, I):

```
-----  
-----  
MAIN MENU  
-----  
-----  
  
- Press 1 to Add Expense  
- Press 2 to Remove Expense  
- Press 3 to Update Expense  
- Press 4 to Display Expense Table  
- Press 5 to Display Yearly/Monthly Expense  
- Press 6 to Display an Expense  
- Press 7 to Quit
```

Login Interface:(Refer: Appendix G)



The image shows a login interface within a light gray rectangular frame. At the top left, the word "Login" is displayed. Below it, there are two input fields: the first is labeled "Username" and the second is labeled "Password". To the right of the "Password" field is a gray button with the text "Login" in white.

Test Plan:

Test ID	Input	Success Criteria ID	Expected Output
1.	Start the program. a. Login with 'user' and 'admin' usernames with provided passwords. b. Try logging with other credentials.	SC1 SC2 SC3	The login interface should pop. On successful login, a login success message should pop with an 'ok' button. On unsuccessful login program should display a warning message and give an option to try again or quit.
2.	On successful login, the program should display the main menu to the user. a. Input in the user interface any character or integer other than menu options	SC4	MENU DISPLAY 1. Add an expense 2. Remove an expense 3. Update an expense 4. Display Expense Table 5. Display the total number of expenses per month /year. 6. Display an Expense 7. Exit the Program The program should do data validation and display an appropriate message to user and guide the user.
3.	Enter the expense using option 1: 'Add an expense'.	SC5 SC6, SC7, SC8, SC9, SC10, SC12, SC14, SC15, SC17 SC18	Add an expense by the guidance of the program. The date format should be dd/mm/yyyy. The program will request the user as to whether the expense subscription requires monthly or yearly payment. Check the expense table using option "4" once whether the entry has been added correctly. Program should return to Main Menu. Repeat test 6 to see multiple expenses are added and stored.

4.	Use option 2: 'Remove an expense'.	SC13 SC11 SC12 SC16	Check the expense table for the expense the user wants to remove, use option 4 in menu.
5.	Use option 3: 'Update an expense'.	SC14 SC7	Check the expense table for changes in expense as user wanted, use option 4 in the menu.
6.	Use option 4: 'Display Expense Table'	SC16	The program should display all expenses in table format.
7.	Select option 5: 'Display Yearly/Monthly expense'.	SC15	The program should calculate the total amount of expenses being paid in a year and each month.
8.	Select option 6: 'Display an expense'	SC20	The program should display an individual expense record based on user input
9.	Option 7 should exit the program	SC18	The program should exit
10	Cross check MySQL RDBMS directly without the software.	SC19	To see whether all the data is still intact after exiting the software.